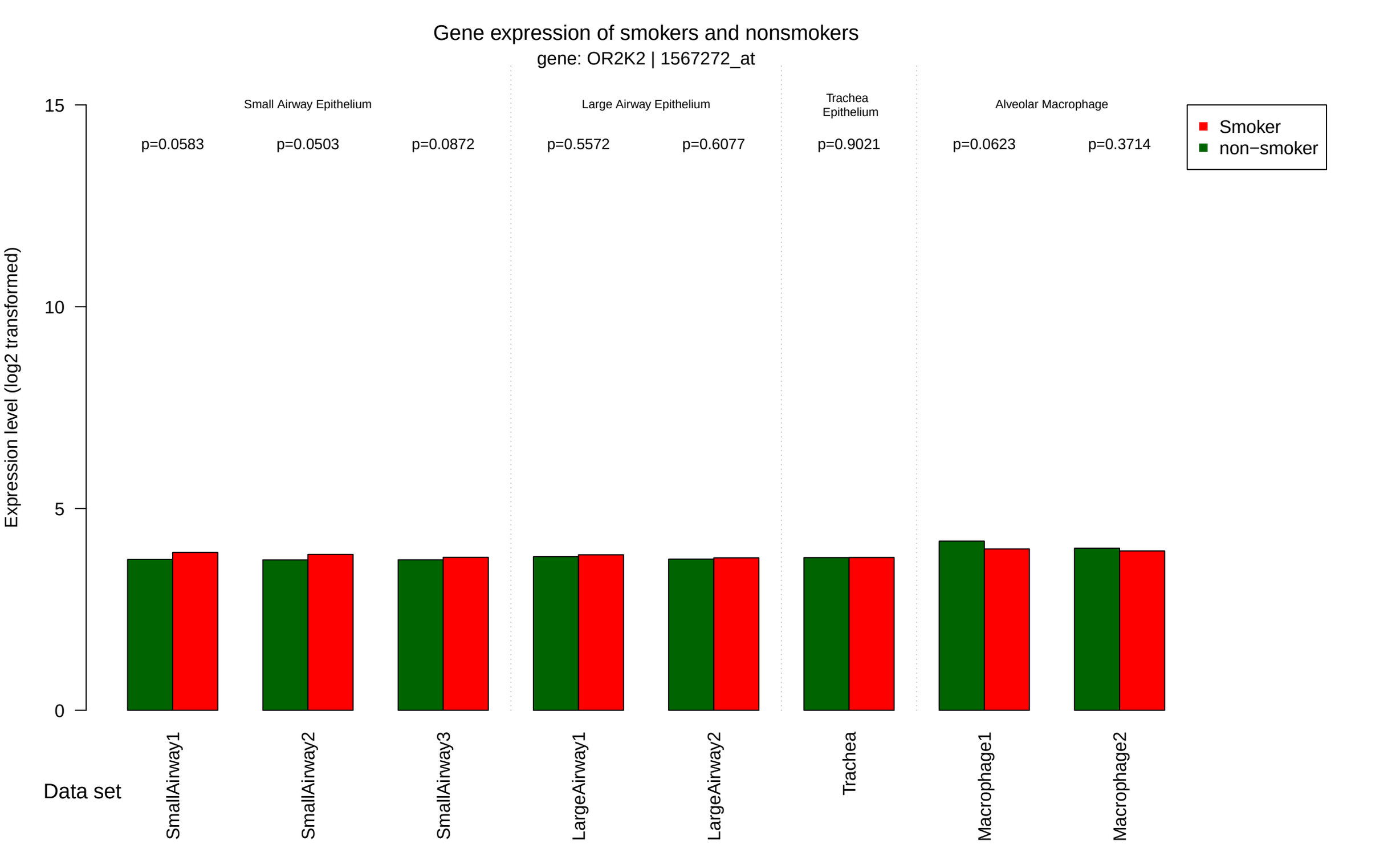
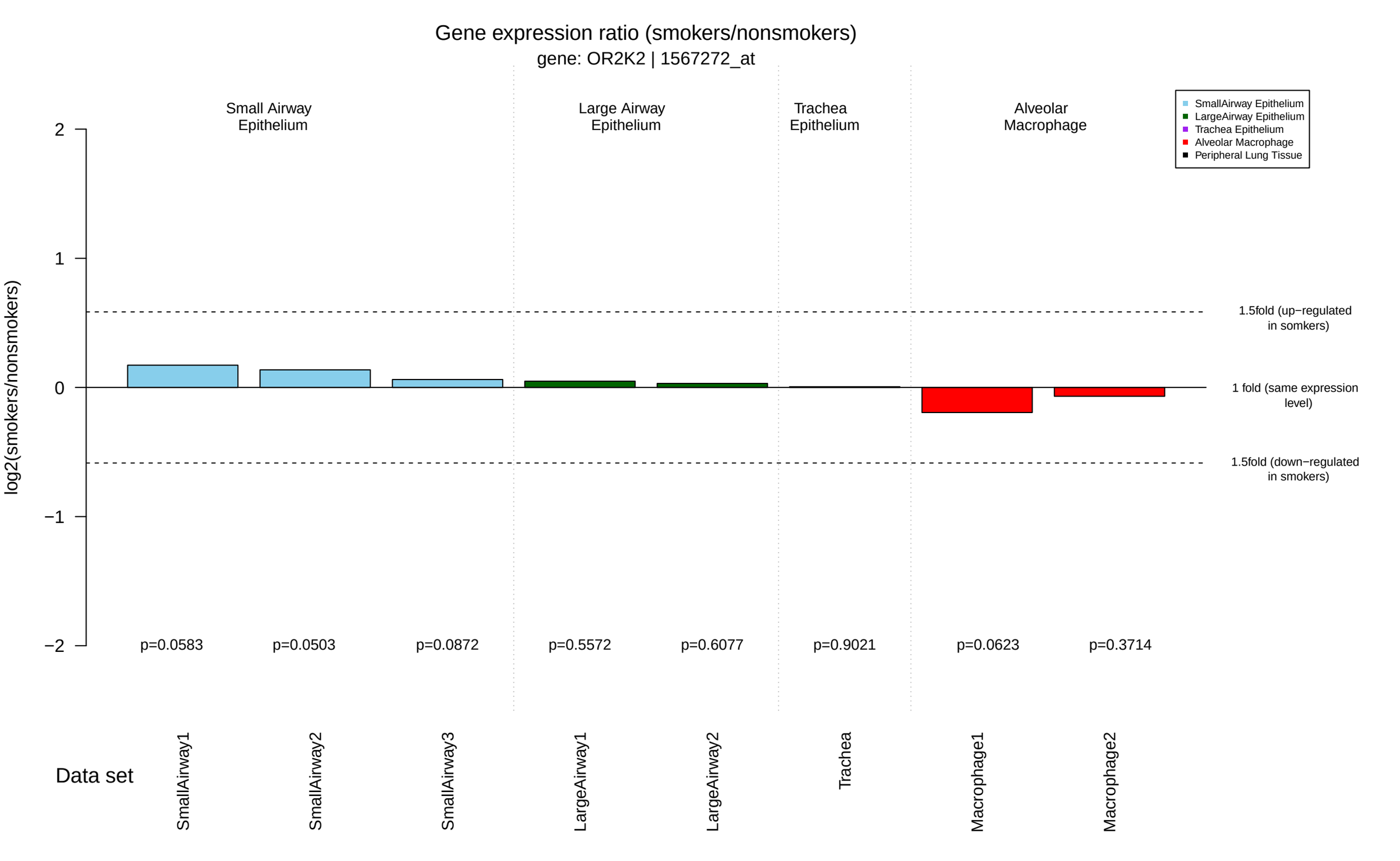
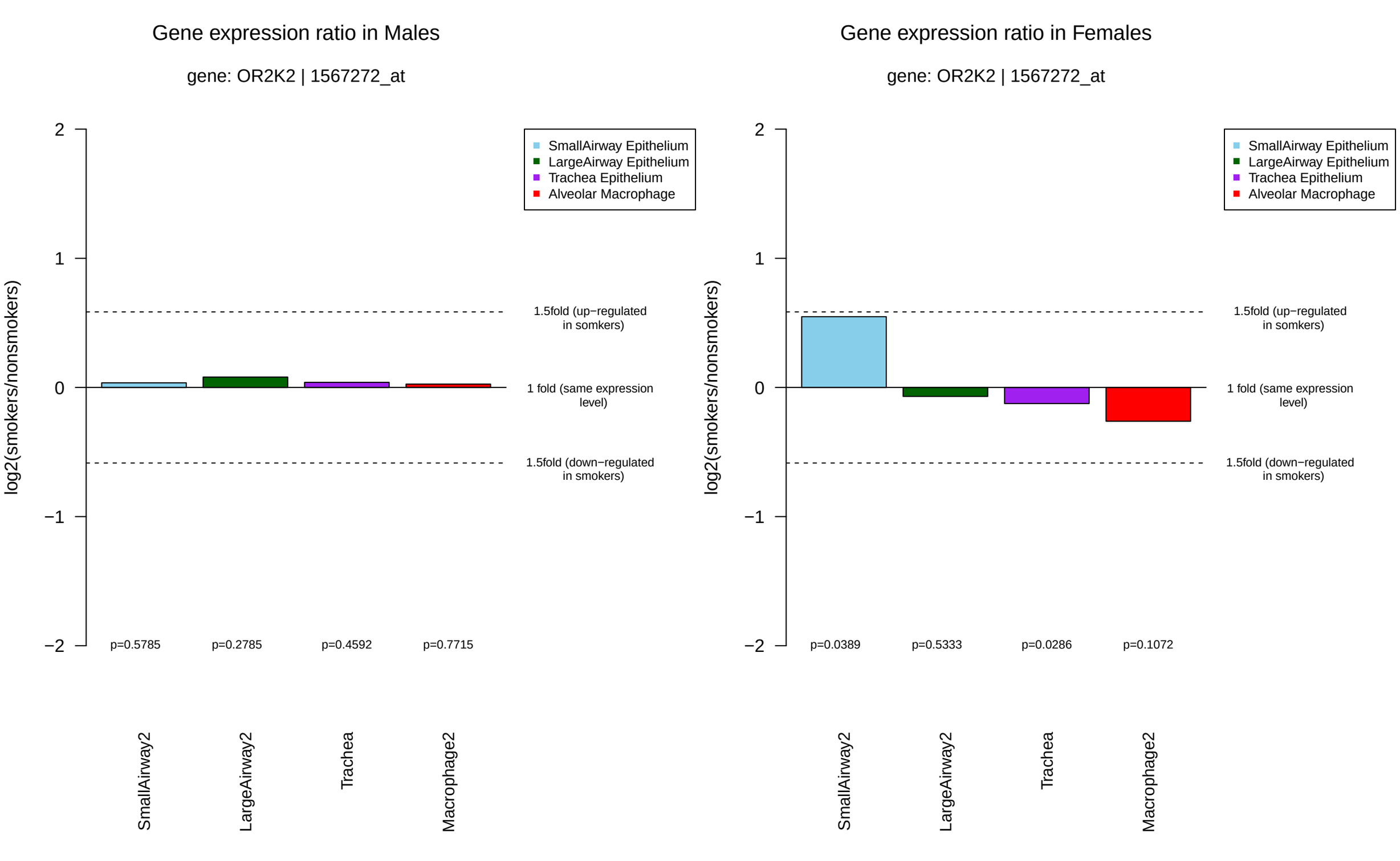
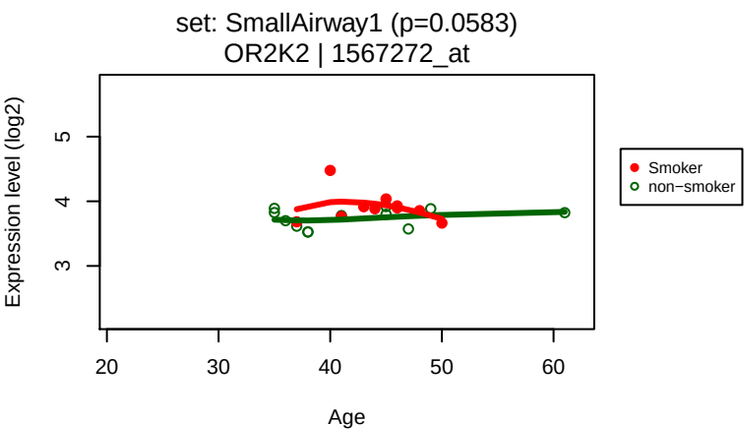
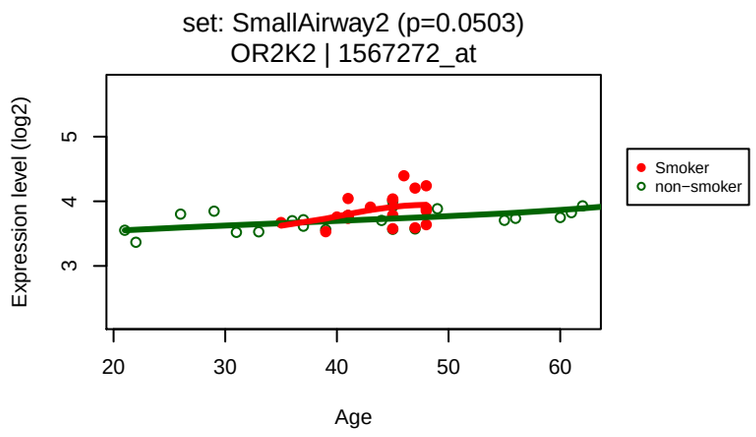
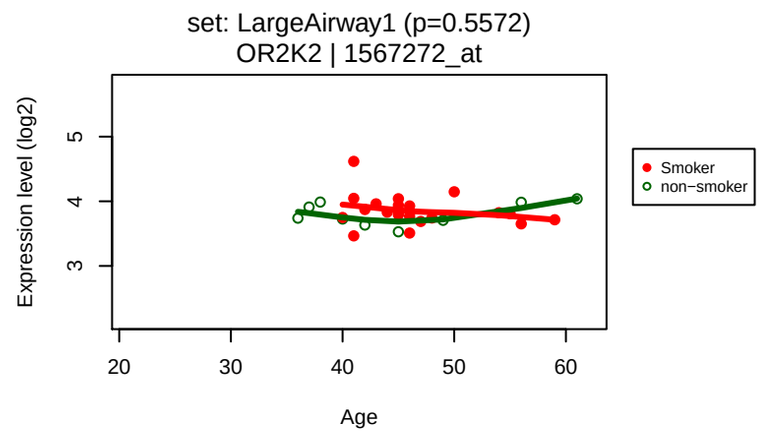

Gene expression of smokers and nonsmokers
gene: OR2K2 | 1567272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0583
p=0.0503
p=0.0872
p=0.5572
p=0.6077
p=0.9021
p=0.0623
p=0.3714
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

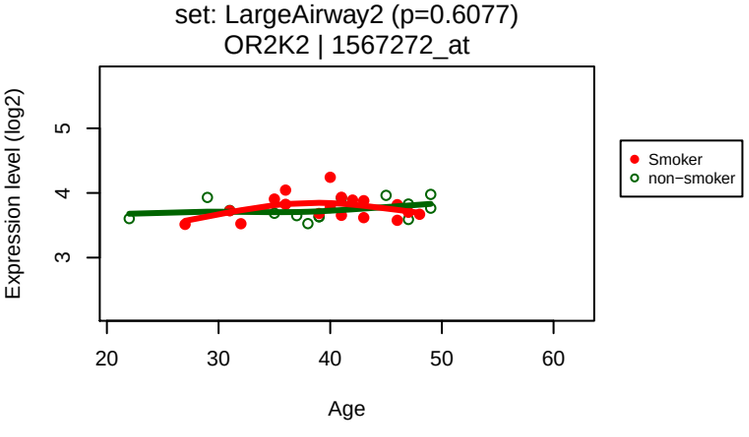

Gene expression ratio (smokers/nonsmokers)
gene: OR2K2 | 1567272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0583
p=0.0503
p=0.0872
p=0.5572
p=0.6077
p=0.9021
p=0.0623
p=0.3714
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

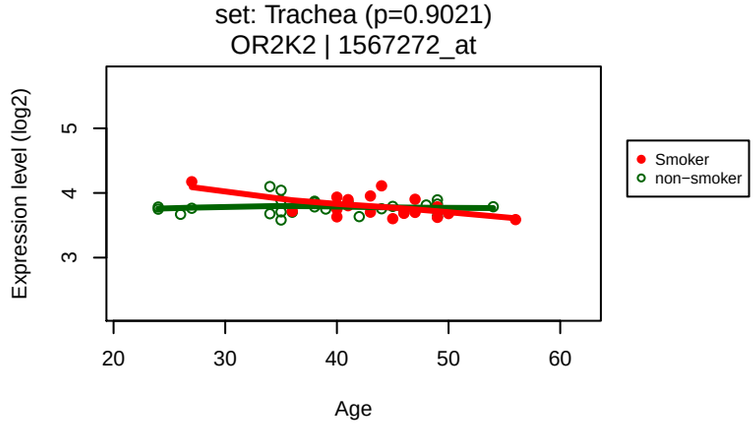

Gene expression ratio in Males
gene: OR2K2 | 1567272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5785
p=0.2785
p=0.4592
p=0.7715
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR2K2 | 1567272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0389
p=0.5333
p=0.0286
p=0.1072
SmallAirway2
LargeAirway2
Trachea
Macrophage2

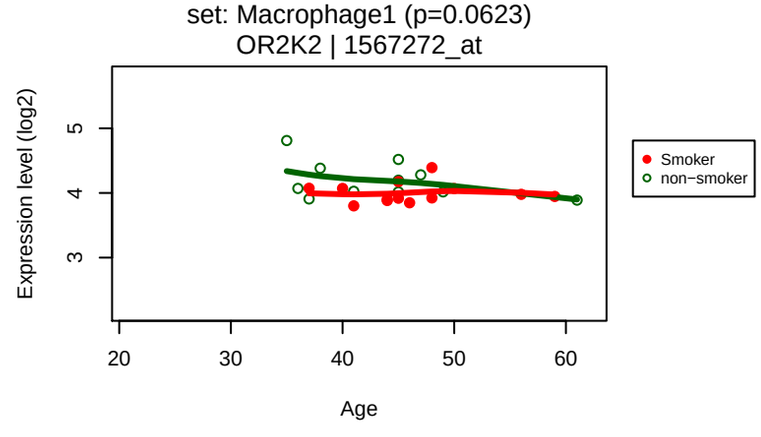

set: SmallAirway1 (p=0.0583)
OR2K2 | 1567272_at
Expression level (log2)
Age
Smoker
non−smoker

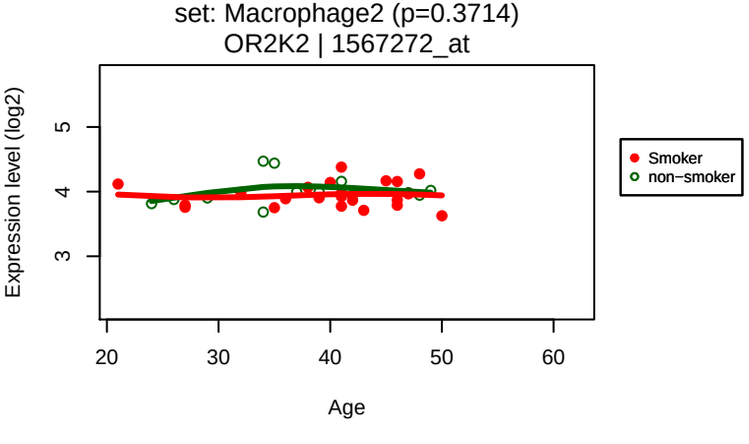

set: SmallAirway2 (p=0.0503)
OR2K2 | 1567272_at
Expression level (log2)
Age
Smoker
non−smoker

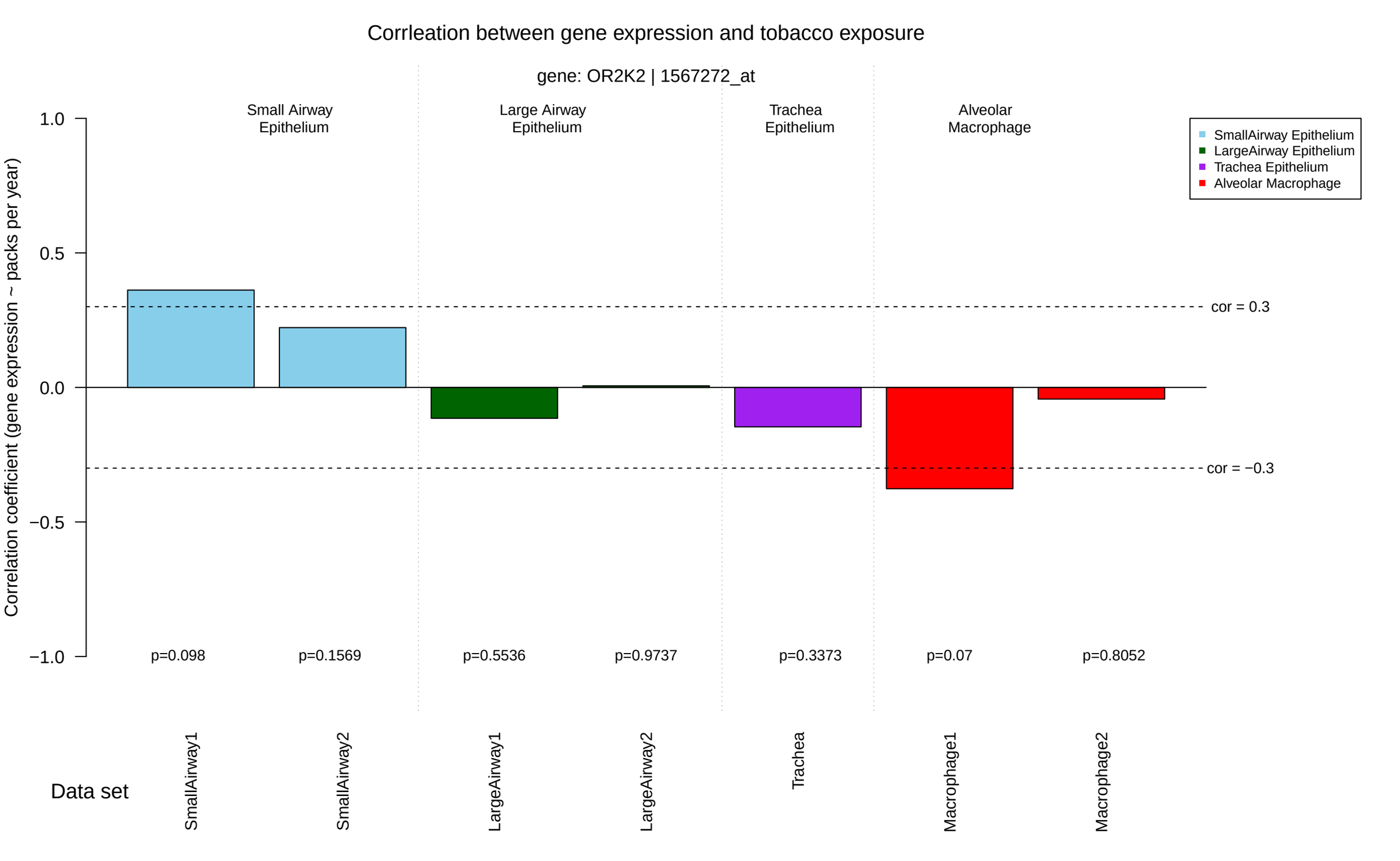

set: LargeAirway1 (p=0.5572)
OR2K2 | 1567272_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6077)
OR2K2 | 1567272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9021)
OR2K2 | 1567272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0623)
OR2K2 | 1567272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3714)
OR2K2 | 1567272_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2K2 | 1567272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.098
p=0.1569
p=0.5536
p=0.9737
p=0.3373
p=0.07
p=0.8052
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2